# AVINASH KUMAR K. M.

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*PhD Scholar, IIT Dharwad · Mechanical Engineering · Medical Image Processing · Computational Geometry · ML/DL*

## RESEARCH INTERESTS

Image-based 3D reconstruction from medical scans · Segmentation algorithms and accuracy characterisation · Morphological analysis of biological structures · Computer-aided geometric design (Bezier, B-splines, NURBS) · Machine learning and deep learning for image analysis · Computational mechanics and FEA

## EDUCATION & RESEARCH EXPERIENCE

**PhD - Mechanical Engineering** | *IIT Dharwad* 2020 – Present

Thesis: Case-specific 3D models of arbitrarily complex branched topologies for modelling morphology

- Developed end-to-end CT/micro-CT to 3D mesh pipeline: multi-algorithm segmentation → Marching Cubes reconstruction → mesh smoothing → morphological analysis
- Quantify mesh smoothing behaviour using Laplace-Beltrami eigenvalue spectrum and a geometry-adaptive threshold values
- Characterised statistical distributions of voxel-resolution-induced surface errors across 9 geometries; generalized gamma identified as best-fit via MLE + AIC/BIC/CvM
- Validated reconstruction pipeline on micro-CT printed phantoms (sphere, facemask, AAA); studied effect of class imbalance and alignment on Dice, Jaccard, Hausdorff metrics
- Built a PyQt-based desktop application for batch morphological analysis of vascular networks across retinal and cerebral domains
- Co-supervised B.Tech project on morphological analysis of femur segmented from clinical CT data
- Implemented transfer learning for multi-class areca nut image classification

Supervisor: Dr. Samarth S. Raut, Assistant Professor, Dept. MMAE, IIT Dharwad | CGPA: 8.47

**MTech - Machine Design** | *RV College of Engineering, Bengaluru (VTU)* 2016 – 2018

Thesis: Numerical investigation of hardness and residual stresses in an induction hardened rocker arm bearing shaft - GTCI, SKF Bengaluru

- The project dealt with simulation of the induction hardening process
- The heat transfer simulation was carried out using the CFD tool STARCCM+, followed using CAM tool DEFORM to compute the residual stresses due to quenching

Supervisor: Dr. Nataraj J. R., Associate Professor, RVCE | CGPA: 8.97

**B.E. — Mechanical Engineering** | *The National Institute of Engineering, Mysuru (VTU)* 2010 – 2014

Thesis: Design of control system to delay stall using active flow control at subsonic flows - CTFD, CSIR-NAL Bengaluru

- Validation of a turbulence CFD model on NACA0012 aerofoil.
- Inclusion of one of the many active flow-control methods available i.e., synthetic jets.
- Simulating an array of piezo resistive sensors to pick up stalling pressure prevalent over the aerofoil.
- Design of a control system to simulate synthetic jets and optimize speed and angle of mass ejection

Supervisor: Dr. T. N. Shridhar, Professor, NIE Mysuru | CGPA: 9.03

## TEACHING EXPERIENCE

**Teaching Assistant** | *IIT Dharwad* 2020 – Present (10 Semesters)

**Laboratory Courses**

- Kinematics and Dynamics of Machinery Lab - demonstrated experiments, guided students through balancing of engines and bearings; evaluated lab reports
- Strength of Materials Lab - conducted sessions on curved beams, membranes, photoelasticity, hardness testing, tensile testing, torsion, and bending

- Engineering Graphics and Solid Modelling Lab - taught 2D/3D drawing conventions, CAD fundamentals (SolidWorks); graded drawings and design projects

**Theory Course**

- Introduction to Programming - delivered lectures to B.Tech and MTech students covering C++, Python, and MATLAB from first principles; designed beginner-level problem sets and assessments

**Courses Capable of Teaching (UG / PG level)**

Strength of Materials · Engineering Graphics · Kinematics & Dynamics of Machinery · Machine Design · Finite Element Analysis · Computational Methods in Engineering · Introduction to Programming (C++/Python/MATLAB) · Computer-Aided Geometric Design · Medical Image Processing (elective) · Machine Learning for Engineers (elective)

## PROFESSIONAL EXPERIENCE

**Research Assistant** | *Raman Research Institute, Bengaluru* Jul 2018 – Aug 2020

Project: RRI Efficient Linear Array Imager (ELI) - novel method to image the galactic plane at 10–30 GHz

- Structural design, FEA (ANSYS Workbench), and 3D modelling (Autodesk Inventor) for backup structures of ELI radio telescope array
- Managed fabrication vendor pipeline; resource and financial oversight at Gauribidanur Observatory, Karnataka

**Operations Engineer — Bar Rod Mill** | *BMM ISPAT LTD., Hosapete* Oct 2014 – Aug 2016

- Supervision and commissioning of rolling stands, cooling bed, stacker, and handling systems
- Process parameter optimisation to maximise production of rebars, rounds, and round corner squares (RCS)

## PUBLICATIONS

**Journal Papers**

- Avinash Kumar K M, Samarth S. Raut, *"Statistical distributions of voxel grid resolution induced surface deviations in marching cube-based 3D reconstructed regular smooth geometries,"* submitted, Computer Aided Design
- Avinash Kumar K M, Samarth S. Raut, *"Influence of Geometry, Class Imbalance and Alignment on Reconstruction Accuracy — A Micro-CT Phantom-Based Evaluation,"* preprint arXiv:2602.07658 (2026)
- Prakash Jayabal, Karolinekersin Enoch, Avinash Kumar K M, Samarth S. Raut, Suneel Rallapalli, Vijay Singh Parihar, Minna Kellomäki, Murugan Ramalingam, Santosh Mathapati, *"3D Bioprinting in Disease Modeling: Integrating Mechanical and Clinical Cues for Physiologically Relevant Systems,"* under preparation

**Conference Papers & Presentations**

- Avinash Kumar K M, Samarth S. Raut, *"Error assessment in image-based 3D reconstruction and 3D printing,"* 30th Congress of the European Society of Biomechanics (ESB 2025), Zurich, Switzerland
- Avinash Kumar K M, Aishwary Singh, D. Narasimha, Rahul Maurya, Samarth Raut, *"Transfer learning based multi-class areca nut image classification under uncontrolled lighting on a conveyor system for automated sorting,"* IEEE Conference on Engineering Informatics (ICEI 2024)
- Avinash Kumar K M, Hemantha Manjunatha, Aashish V Bhat, Bhagyashree Kulkarni, *"Active flow control using Synthetic jets and neural network,"* 16th Annual CFD Symposium, CSIR-NAL, Bengaluru, 2014

## AWARDS & RECOGNITION

- 2nd Best Poster Award for the ESB paper - ANRF-INAE Conclave on Atmanirbhar Technologies: Engineering a Secure Future, "Indigenous Technology Development" theme (2025)
- Qualified GATE (Graduate Aptitude Test in Engineering) — Mechanical Engineering (2014, 2016, 2020)

## TECHNICAL SKILLS

**Programming:** C++, Python, MATLAB

**Medical Imaging:** DICOM, NIfTI, Pydicom, ITK/SimpleITK, 3D Slicer, TotalSegmentator, Marching Cubes, micro-CT reconstruction

**ML/DL:** PyTorch, TensorFlow, scikit-learn, transfer learning, CNN architectures, CUDA GPU programming

**Mesh & Geometry:** Laplacian/Taubin/HC-Laplacian smoothing, Laplace-Beltrami spectral analysis, cotangent Laplacian, ICP registration

**Simulation & CAD:** ANSYS Workbench (FEA), SolidWorks, Autodesk Inventor

**Documentation:** LaTeX, MS Office

## REFERENCES

**Dr. Samarth S. Raut**
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**Dr. Amar Keshav Gaonkar**
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